



Day 0 Speaker Notes

Slide 2 Introduction

- Introduce yourself
- Ryan Pelchat, Masters Student in Computer Science
- Last couple years competed in ICPC
- Free time, read the Expanse, tinker with hardware+software (Jellyfin, FreeNAS,)

Slide 3 What is ICPC?

- ICPC is the International Collegiate Programming Competition
- Teams of 3 have 5 hours to solve 13 problems and 1 computer
- All the paper that you can bring, no internet allowed
- Secret tests, time penalty if you submit a wrong solution
- Covers wide breadth of Computer Science
- Requires: Teamwork, Knowledge, and Practice
- Will test how well you can:
 - create data structures
 - execute algorithms
 - and optimize your code
- Why?
- Greater mastery of programming and Computer Science
- Help you with coding based interviews
- It's fun!

Slides 4 Format

- discuss previous day's contest (vote on what problems to talk about)
- instruction of new material
- lunch
- practice contest
- Upload your solutions to eclass so that we can see your progress

- Read the readings posted to eclass before the next day

Slide 5-9 Roadmap

Slide 10 Resources

- Competitive Programming 4
- Python Programming, Introduction to Computer Science

Slide 11 How to be successful in ICPC

- Theory
 - Use CP4 textbook as a roadmap of the skills required in ICPC
 - Use many resources to learn!
 - Data Structures, Algorithms textbooks
 - Lectures from other universities
- Practice!
- Do the problems in CP4
- Teamwork
 - You have 3 ppl, important to split the work efficiently
 - You have to practice together to get to know each other's strengths and weaknesses. To know how each other think.
 - Strategize! Make a plan on how to classify your problems and distribute them.
 - Pair programming (Ask who has done so and explain what it is)

Slide 12 Example Strategy

- Start:
 - Every member reads different problems
 - Rank each problem, Easy, Medium, Hard
 - Easy -> confident can solve on your own
 - Medium -> somewhat confident, need someone to talk it through, maybe pair program

- Hard -> might need all 3 members, unsure, keep for last
- Classify the problem type
 - Dynamic Programming, Minimum Spanning Tree, Binary Search, etc...
 - This is to be able to refer problems to each other
- Interrupt:
 - While reading, if a member ranks an easy, attempt to solve it right away
- Computer Usage
 - First solve on paper to verify your solution, unless it is an easy and no one is at the computer
 - Have 2 signs
 - “quick test/print” for tasks under 1 minute
 - “ready to code” for longer coding tasks
 - This allows the person at the computer to dictate when they stop
 - For a quick second or if they are not making progress (debugging and back to paper solving)

Let's get to know each other

- Go around the room and introduce ourselves, and why you are interested in ICPC